CONCRETE SERVING TRAY

Step-by-Step DIY Guide

Cast a Stunning Textured Concrete Tray with Natural Imprints

The finished concrete serving tray — styled with books, coffee, and natural objects

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About This Project

This concrete serving tray is one of the most versatile and impressive DIY home decor projects you can make. By pressing natural objects — leaves, pebbles, stones, and terracotta shards — into the surface of wet concrete, you create a tray that is completely unique and tells a story. The concrete captures every texture and detail permanently. Use it as a coffee table centerpiece, a bathroom organizer, a desk tray, or a stunning serving board for entertaining.

Difficulty	Easy
Active time	30–45 minutes
Curing time	24–48 hours
Finished size	~40 x 25 cm (depends on mold size)
Cost estimate	$8–15 USD
Best for	Home decor, gifts, styling props, entertaining
Weight	~1.5–2 kg — feels luxurious and substantial

Materials & Tools

#	Item	Notes
1	Craft concrete or GFRC mix	Fine-grain concrete gives the smoothest finish. Avoid coarse aggregate mixes. GFRC
2	Water	Clean tap water. Mix ratio: approx. 2.5 parts concrete to 1 part water by volume.
3	Rectangular plastic mold / tray	A plastic storage box or photo developing tray ~40x25 cm and 3–4 cm deep works pe
4	Mixing bucket + old spoon	For mixing the concrete. A 5-litre bucket is sufficient.
5	Flat scraper or trowel	To spread and level the concrete evenly in the mold.
6	Natural objects for texturing	Leaves (press-flat varieties like bay leaves, eucalyptus), smooth river pebbles, stone
7	Textured mat (optional)	A silicone texture mat, lace, or rough fabric pressed into the surface creates geometr
8	Nitrile gloves	Essential — wet concrete is alkaline and irritates bare skin.
9	Sandpaper (150–220 grit)	For smoothing the edges and underside after demolding.
10	Felt pads or cork sheet	Glue to the underside of the tray to protect surfaces it sits on.
11	Concrete sealer	One or two coats on all surfaces — protects against stains, moisture, and scratches.
12	Cooking spray or petroleum jelly	Lightly coat the inside of the mold before pouring for easy demolding.

■ ***Mold tip:*** *The cleaner and smoother the inside of your mold, the smoother the top surface of your tray will be. A wipe with petroleum jelly makes demolding effortless and leaves a slightly glossy surface.*

Step-by-Step Instructions

Step 1 — Mix the Concrete

Mixing concrete in a bucket to a smooth, thick-yoghurt consistency

Combine your concrete mix and water in a bucket. The ideal ratio is **2.5 parts dry mix to 1 part water** by volume. Mix vigorously with a spoon or stick for at least 3 minutes until completely smooth with no lumps or dry powder remaining. The consistency should be thick but pourable — like thick yoghurt or oatmeal.

- Add water slowly — slightly too little water is easier to fix than too much.
- Scrape the bottom and sides of the bucket while mixing to incorporate all dry mix.
- Let the mix rest for 2 minutes after mixing, then stir once more — this improves workability.
- A small drop of dish soap in the water helps reduce surface air bubbles.

> ■ ***Safety:*** *Always wear gloves when handling wet concrete. Prolonged contact with bare skin causes irritation and can lead to cement burns.*

Step 2 — Prepare the Mold & Pour

Pouring concrete into the rectangular plastic mold — the tray shape takes form

Lightly coat the inside of your rectangular mold with cooking spray or a thin layer of petroleum jelly — this is your mold release agent. Then pour the concrete mix into the mold in a steady stream, starting from one end and working towards the other.

- Pour to a depth of 2.5–3 cm for a sturdy tray that won't crack under weight.
- Pour slowly to reduce splashing and trapped air.
- Tap the sides of the mold firmly for 30–60 seconds after pouring to release air bubbles.
- Tilt the mold slightly back and forth to help concrete reach all corners evenly.

■ ***Tip:*** *For a lighter tray, fill to only 2 cm depth. For a heavier, more luxurious feel, go to 3–4 cm. Thinner trays cure faster but are more fragile.*

Step 3 — Smooth the Surface

Smoothing the concrete surface flat with the back of the hand — creating an even, level tray

Once poured, use a flat scraper, trowel, or even the back of your gloved hand to smooth the surface of the concrete level. This becomes the *underside* of the tray — but a flat, smooth base means the tray will sit perfectly stable on any surface.

- Work in long, sweeping strokes from one end to the other.
- Check the surface is level by eye — use a spirit level if you want perfection.
- Remove any excess concrete that mounds above the mold rim with a straight edge.
- Don't over-work the surface — too much movement brings water to the top and weakens the mix.

> ■ ***Tip:*** *The smoother and flatter you get this surface, the more stable your finished tray will sit. Take your time on this step.*

Step 4 — Press Texture Objects into the Surface

Arranging and pressing leaves, pebbles, stones, and terracotta pieces into the wet concrete surface

This is the most creative step — and where your tray becomes truly one of a kind. While the concrete is still wet and workable (within 20–30 minutes of pouring), arrange your chosen objects — leaves, stones, pebbles, terracotta shards — on the surface and press them in firmly. They will leave permanent imprints when removed after curing.

- Press each object firmly into the surface so at least half its depth is submerged.
- Flat objects like leaves imprint beautifully — the veins and edges all transfer.
- Round pebbles create perfect circular depressions.
- Terracotta or ceramic shards add irregular organic shapes.
- Arrange freely — close together for a busy, maximalist look, or spaced apart for minimalist elegance.
- You can also press in a textured silicone mat over the entire surface for a repeating geometric pattern.

■ ***Tip:*** *Press the objects in DEEPLY rather than just resting them on top. A deep impression creates a more dramatic, defined texture in the finished tray.*

Step 5 — Cure for 24–48 Hours

Misting the cured tray surface with water — keeping it damp speeds curing and reveals the texture

Leave the concrete undisturbed for **24–48 hours**. The objects you pressed in will stay in place while the concrete hardens around them. Cover the mold loosely with plastic wrap to slow the cure and prevent cracking. In warm weather, lightly mist the exposed surface with water every few hours.

- Do NOT disturb or move the mold during curing — even small vibrations can crack the slab.
- Keep out of direct sun — shade prevents too-fast drying and surface cracking.
- After 24 hours, the concrete will be firm enough to touch — but wait the full 48h for best results.
- In cool weather (below 15°C), give it an extra 12–24 hours before demolding.

> ■ ***Tip:*** *After 24 hours, try gently pressing the side of the mold to test stiffness. If it feels rock solid, you're ready to demold. If there's any flex or softness, wait longer.*

Step 6 — Demold, Remove Objects & Finish

After full curing, flex the plastic mold gently from all sides to break the seal, then turn it upside-down onto a padded surface. The tray should drop out cleanly. Now remove the objects pressed into the surface — their imprints will be permanently captured in the concrete.

- Tap the bottom of the mold if the tray doesn't release immediately.
- Peel leaves carefully — any bits stuck in the concrete can be picked out with a toothpick.
- Rinse the tray with water to clean out any remaining debris from the imprints.
- Sand all edges and the underside with 150-grit then 220-grit sandpaper — wear a mask.

• Wipe clean with a damp cloth and allow to fully dry for 24 hours before sealing.

> ■ ***Tip:*** *Wet-sand the edges and underside for a cleaner result with much less concrete dust. Dip the sandpaper in water as you work.*

Step 7 — Seal & Add Feet

The sealed tray styled as a coffee table centerpiece — books, coffee, and natural objects

Apply 2 coats of concrete sealer to all surfaces of the dry tray, allowing each coat to dry fully between applications (usually 1–2 hours). The sealer protects against water, stains, and scratches — and gives the tray a subtle sheen that brings out the grey tones beautifully.

• Use a foam brush or cloth to apply sealer in thin, even coats.

• Seal the top, sides, and bottom — all surfaces need protection.

• For serving use: choose a food-safe, non-toxic concrete sealer explicitly.

• After the sealer cures (24h), stick 4 self-adhesive felt pads to the corners of the underside.

• The felt feet protect whatever surface the tray sits on from scratches.

> ■ ***Tip:*** *For an extra luxurious look, apply a coat of beeswax polish after sealing and buff with a soft cloth. It deepens the colour and adds a natural sheen.*

The Finished Tray

The finished concrete serving tray — solid, beautiful, and completely handmade

Pro Tips & Variations

★ Use GFRC for a lighter tray	Standard concrete at 3 cm depth is heavy (~2 kg). GFRC (Glass Fiber Reinforced Concrete) produces a tray 30–40% lighter with equal or greater strength — perfect for a tray you'll carry around frequently.
★ Create a geometric pattern with a texture mat	Instead of natural objects, press a silicone baking mat, lace fabric, or a stair nosing grip mat into the wet concrete. The result is a perfectly repeating geometric texture across the entire surface.
★ Colour the concrete	Adding concrete pigment powder creates a coloured tray. Black or charcoal gives a dramatic slate look. White makes a bright, modern piece. Terracotta oxide creates a warm, earthy tone.
★ Make a two-tone tray	Pour a thin first layer in one colour, let it firm slightly (about 1 hour), then pour a second layer in a different colour on top. When demolded, the tray face shows the first colour and the back shows the second.
★ Add handles	Before the concrete fully cures (~3–4 hours after pouring), push two lengths of thick rope, leather cord, or copper pipe into each end of the tray. When cured, these become permanent handles — both functional and beautiful.
★ Embed objects permanently	Instead of removing the pressed objects after curing, leave them embedded in the concrete. Pebbles, glass gems, or tiles pressed halfway in become permanent decorative elements.
★ Perfect as a bathroom organizer	Seal thoroughly and use to organize soap, cotton pads, a candle, and small bottles. The weight of concrete keeps it perfectly still even in a busy bathroom.
★ Create a set	Make a large tray and two smaller matching coasters from the same pour. Use the same objects for texturing — they make a cohesive, professional-looking set that sells well at craft fairs ($35–80 depending on size).

Troubleshooting

✗ Problem: Surface has lots of pinholes and bubbles

✓ **Solution:** Tap the mold more vigorously and for longer after pouring (60+ seconds of firm tapping). Also try a slightly wetter mix and add a drop of dish soap to the mixing water.

✗ Problem: Tray cracked during curing

✓ **Solution:** Dried too fast (sun/wind) or the mix was too wet. Cover during curing and keep in shade. Hairline cracks can be filled with fresh concrete slurry and sanded once dry.

✗ Problem: Tray stuck in the mold

✓ **Solution:** Didn't use enough mold release, or demolded too early. Flex the mold walls firmly in all directions. If still stuck, try placing in a warm water bath for 10 minutes to soften the plastic mold slightly.

✗ Problem: Imprints are too shallow / not visible

✓ **Solution:** The objects weren't pressed in deeply enough, or the mix was too stiff. Next time, press objects at least halfway into the concrete and use a slightly wetter mix.

✗ Problem: Tray is uneven / wobbles

✓ **Solution:** The surface wasn't levelled properly before curing. Sand the underside flat with coarse sandpaper on a flat surface, checking frequently until stable. Or add self-adhesive rubber feet to even out the base.

✗ Problem: Sealer is leaving streaks or white marks

✓ **Solution:** Applied too thickly or in cold/humid conditions. Wait for a warm dry day (above 15°C). Apply thinner coats and buff with a cloth immediately after applying each coat.

Thank you for your purchase!

Every concrete tray you make is completely unique — no two pours, no two arrangements of objects, and no two finishes are ever exactly the same. Enjoy making something truly one of a kind.

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